



WELCOME

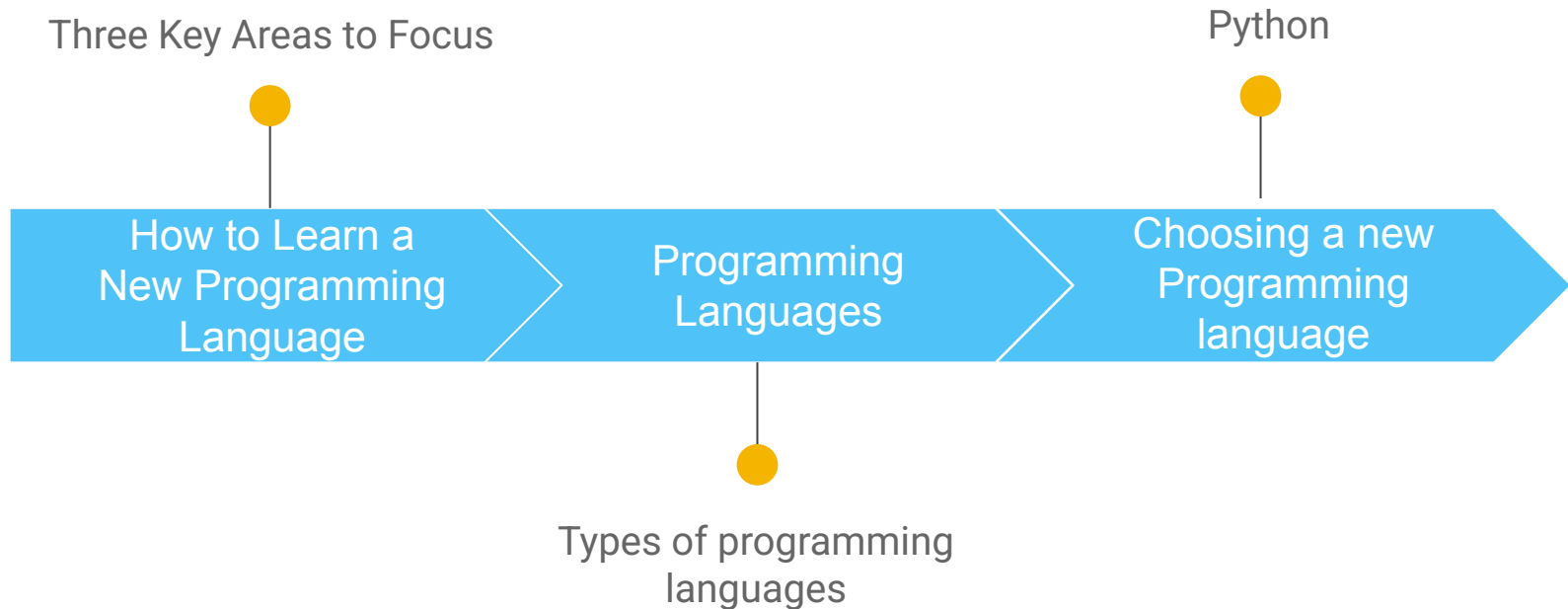




How to Learn a New Programming Language in Record Time

Agenda for Today's Session

How to Learn a New Programming Language in Record Time



How to Learn a New Programming Language



Analogy

Driving a Car

Let's say,

You know how to drive a **Mahindra**



Driving a Car

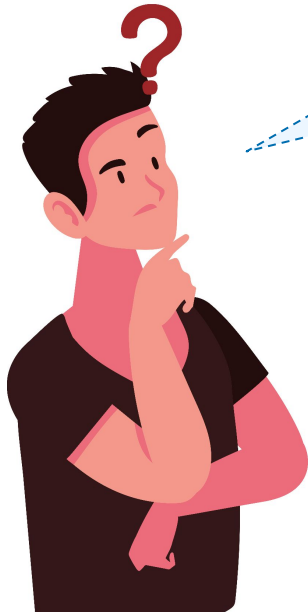
New Car

Now, **You have to Drive a Tata**



Driving a Car

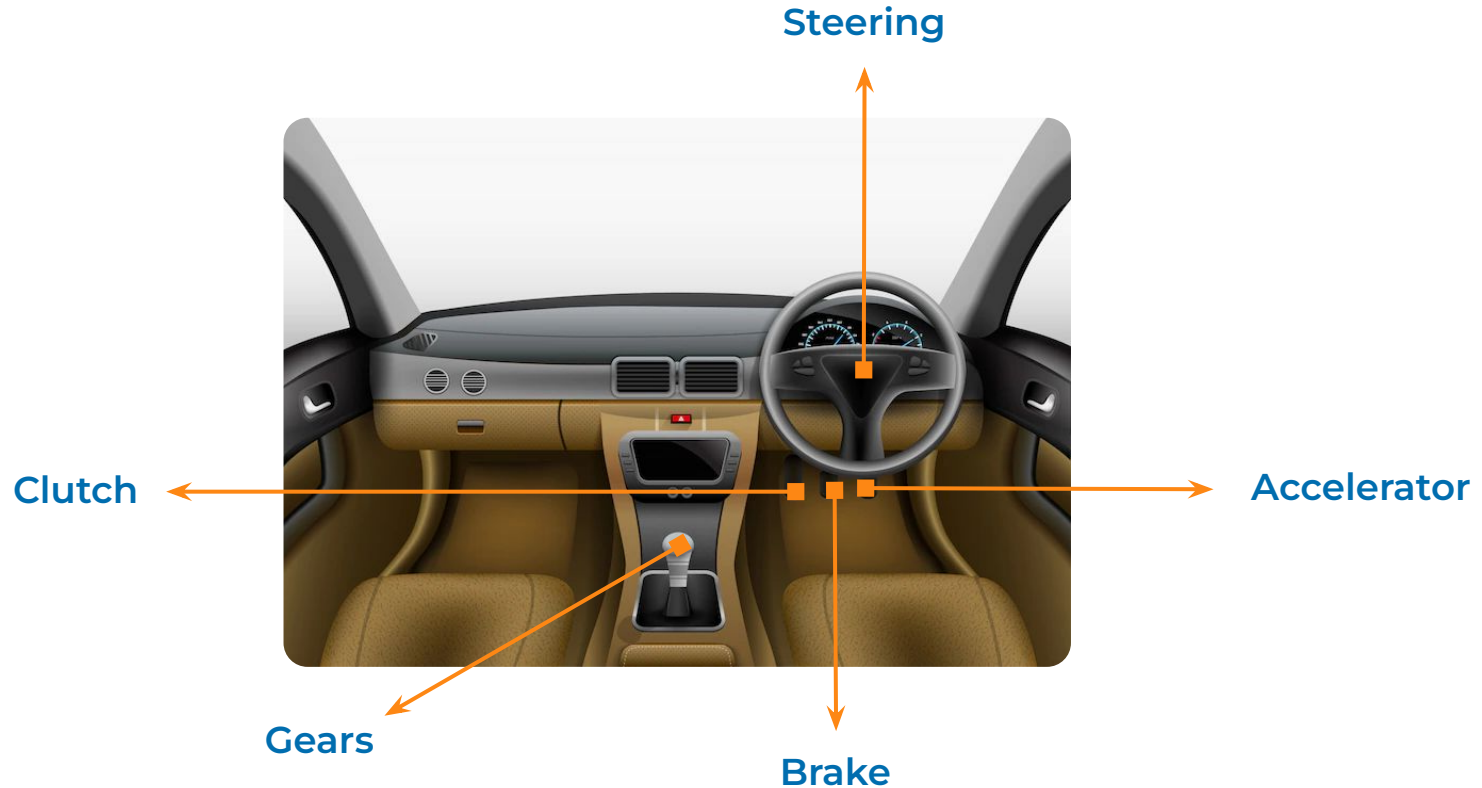
New Car



Do you have to learn driving from the scratch?

Driving a Car

Common Aspects



Driving a Car

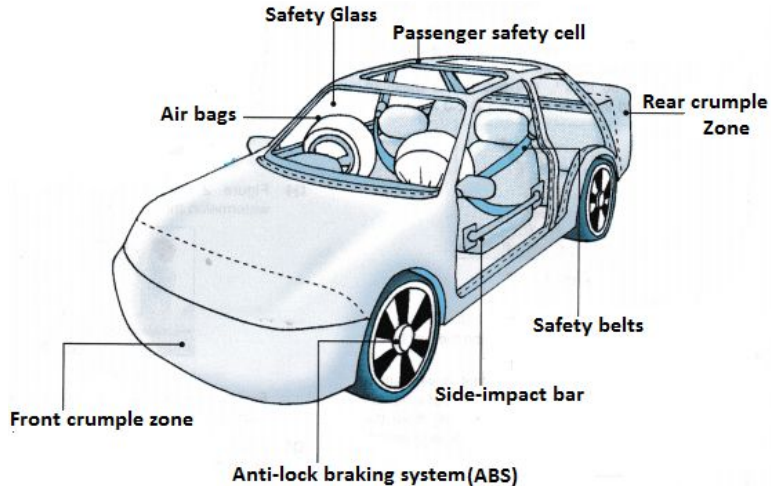
Mahindra vs Tata



What are the differences between the two cars?

Mahindra vs Tata

Possible Differences



Acceleration / Speed



Seating Capacity



Diesel / Petrol



Engine Power



Wipers



Driving a Car

New Car

- ❓ Does driving a new car require a different set of fundamental skills?



Example

Person Applying for a Driver Job



Does one need to specify specific cars, like **Mahindra, Tata, Honda**, etc, when applying for a driving job?



Driving a Car

Learning a New Car

How much time is needed to learn how to drive different cars?

Mahindra



Tata



Honda



Learning a New Car

Time Required

How much time is needed to learn how to drive different cars?

Mahindra



~ 30 Days



Tata



~ 5 Hours



Honda



~ 1 Hour

Driving a New Car

Time Required to Adjust to a New Car

It usually only takes a **few minutes** to **adjust** to a new car



Comparison with Driving a New Car

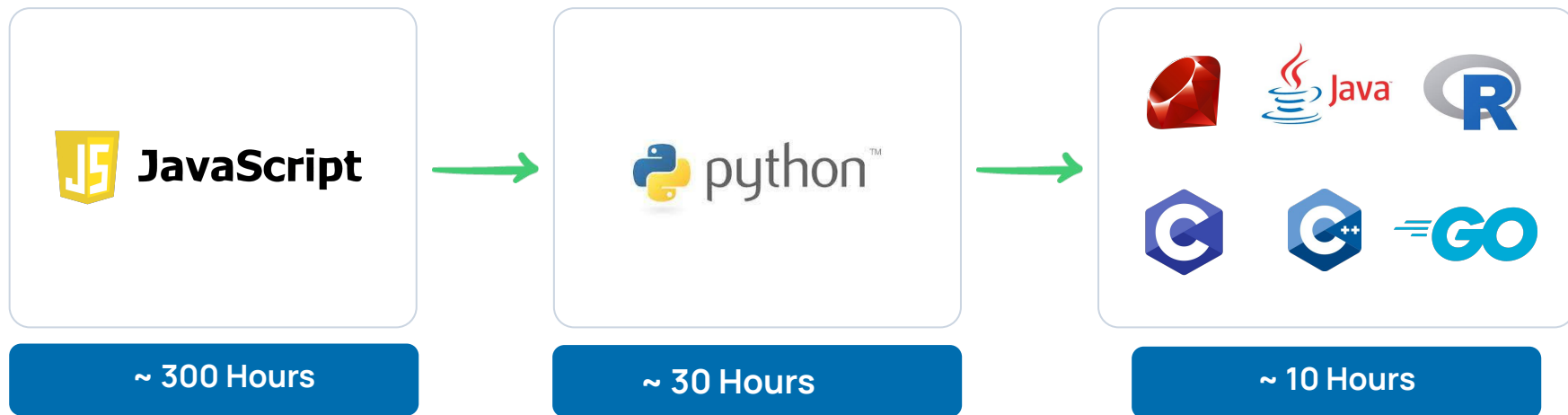
Learning a New Programming Language

- ✓ Learning a **new** programming language is **similar to** learning to drive a new car
- ✓ A **strong foundation** in **one** programming language can make it **easier** to **learn other languages**



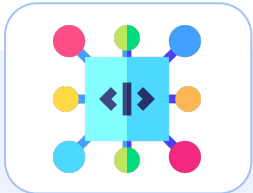
Learning a New Programming Language

Time Required

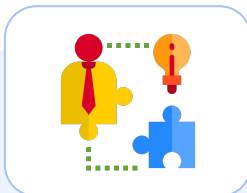


Three Key Areas

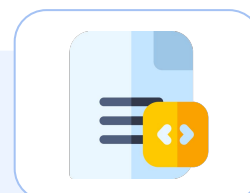
*To become proficient in a programming language,
there are three key areas to focus on*



Programming
Abstractions



Problem
Solving Approach



Syntax

Three Key Areas to Focus

Programming Abstractions



- ✓ Steering
- ✓ Brake
- ✓ Accelerator
- ✓ Clutch
- ✓ Gears ...etc

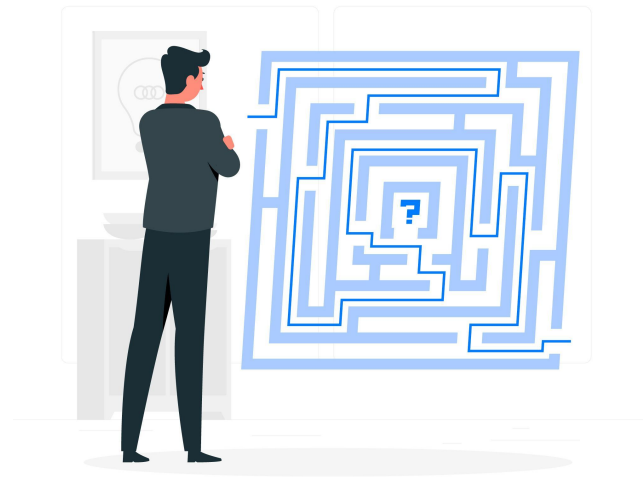


- ✓ Variables
- ✓ If Else Statements
- ✓ For & While Loops
- ✓ Functions
- ✓ Classes ...etc

Three Key Areas to Focus

Problem-Solving Approach

- ✓ Across all programming languages, the **problem-solving** approach is the **same**
- ✓ When you **master** it in **one language**, you can **apply** it to **others**



Three Key Areas to Focus

Syntax



- ✓ The **main difference** between programming languages is **Syntax**, which can be learned in just a **few hours**
- ✓ After that, you just need to **practice** more to become proficient in the language

Choosing a New Programming Language to Learn



Example

Buying a New Car



? What factors do you consider when choosing a car to buy?

Buying a New Car

Considerations



Sports Car?



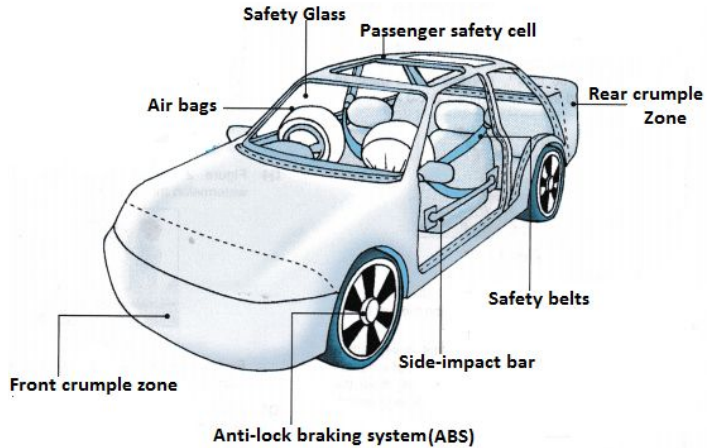
To use in Cities?



Off Road?

Buying a New Car

Considerations



Mileage



Economy



Speed



Seating Capacity



Buying a New Car

Considerations



It is always based on our needs, **correct?**

Choosing a new programming language is **similar** to **buying** a **new car**

Choosing a New Programming Language

Considerations

Globally, there are around **700** programming languages



Types of Programming Languages

There are many ways to **categorize programming languages**, but they **typically fall into**

- ▶ Procedural
- ▶ Object-Oriented
- ▶ Functional
- ▶ Scripting

...and more



Procedural Programming Languages

Procedural Programming Languages use **set of instructions** and specifies **“what to do”** and **“how to do”**



Procedural Programming Languages



Procedural programming languages have **more similarities** compared to other types of programming languages

Object-Oriented Programming Languages



- ✓ Programming Languages that are **designed to support object-oriented** programming
- ✓ Uses **objects** to represent real-world entities and their interactions

Choosing Programming Languages

Companies



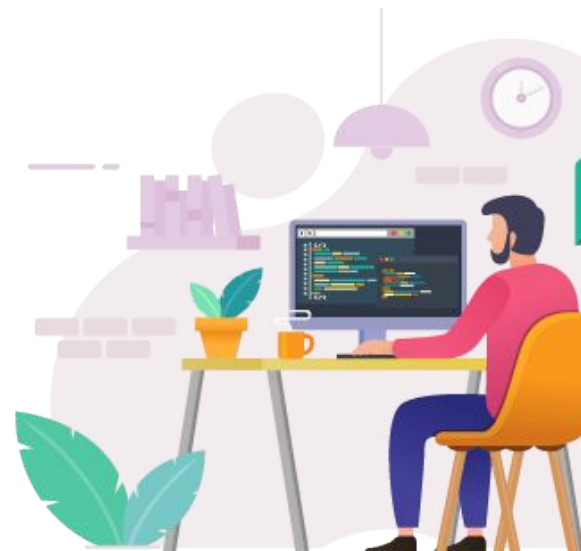
Companies generally **choose**

Programming Languages based on their
specific needs and **requirements**

Programmers

Choosing Programming Languages

- ✓ **Programming** can be a **hobby** for some
- ✓ They choose **programming** languages as a way to **develop new things** and bring their **ideas to life**



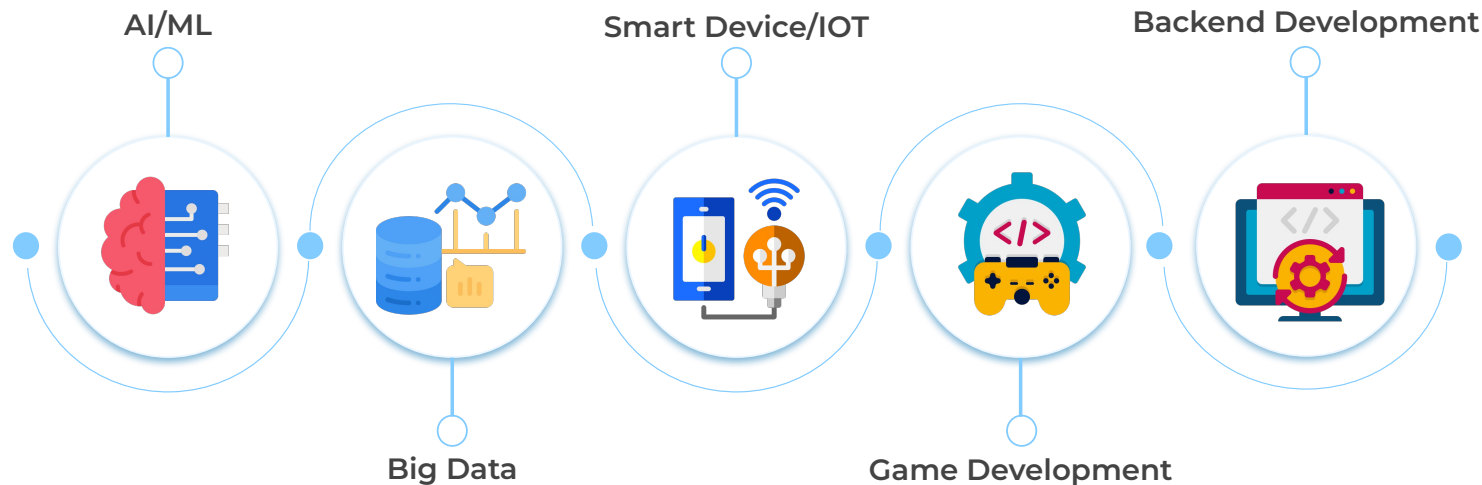
Let's Look at a New Programming Language



Why Python?

Versatile Language

Has applications in almost every field



Why Python?

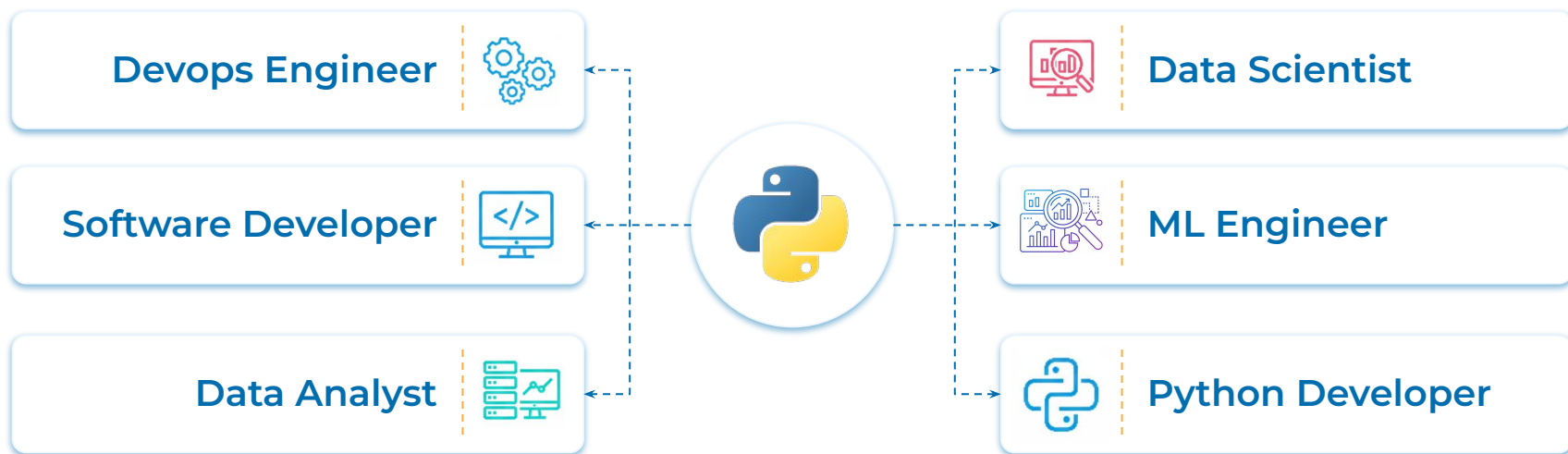
Popular at Top Tech Companies

Major Tech Companies use Python extensively



Why Python?

Plenty of Opportunities



Python Developer - Global

Huge Salaries

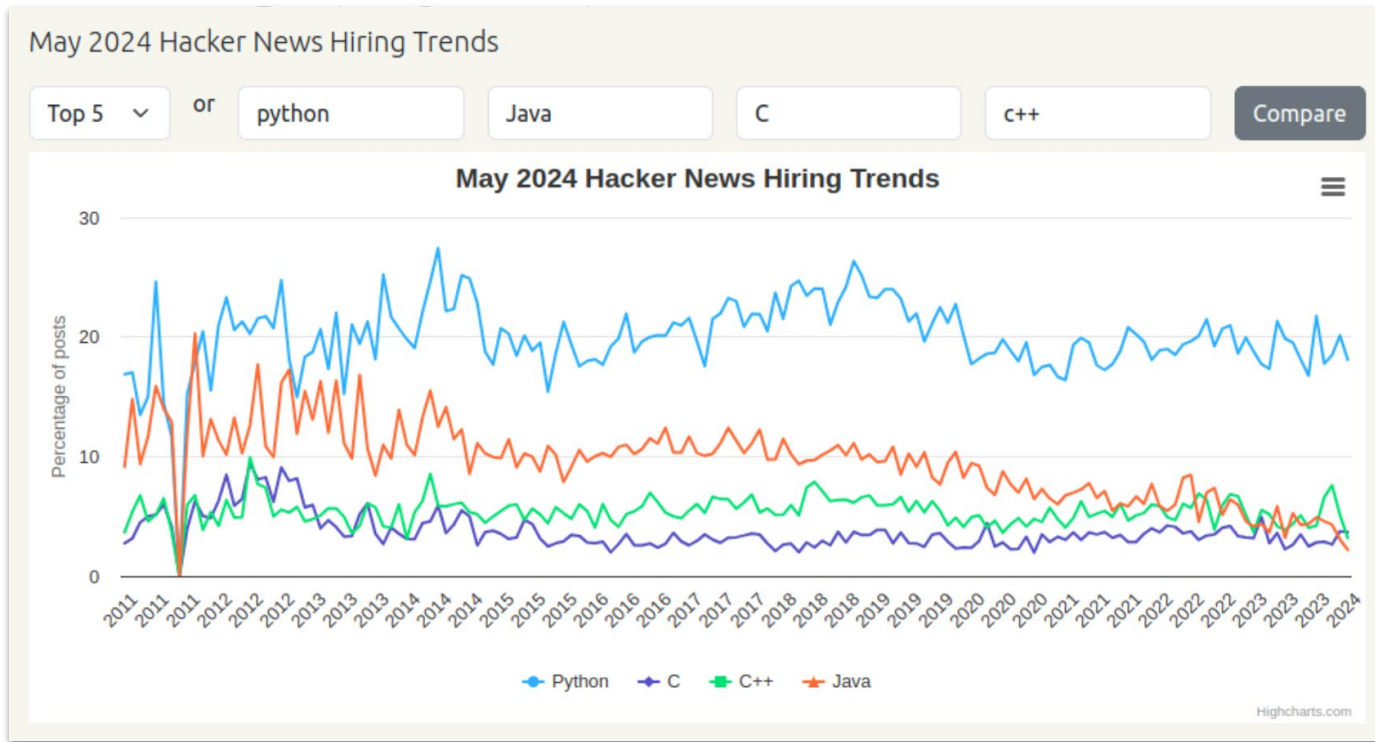
Tech companies like Oracle, Amazon, Google, Microsoft **offer competitive salaries** in both India and the US



Source: [PayScale](#)

Why Python?

Demand for Python

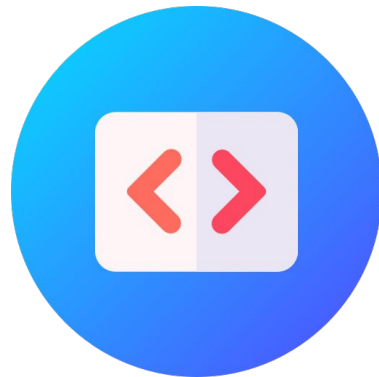


Why Python?

Easy to Learn



Simple Syntax



Minimal Code

Why Python ?

Example

Javascript

```
console.log("Hello World!");
```

Python

```
print("Hello World!")
```

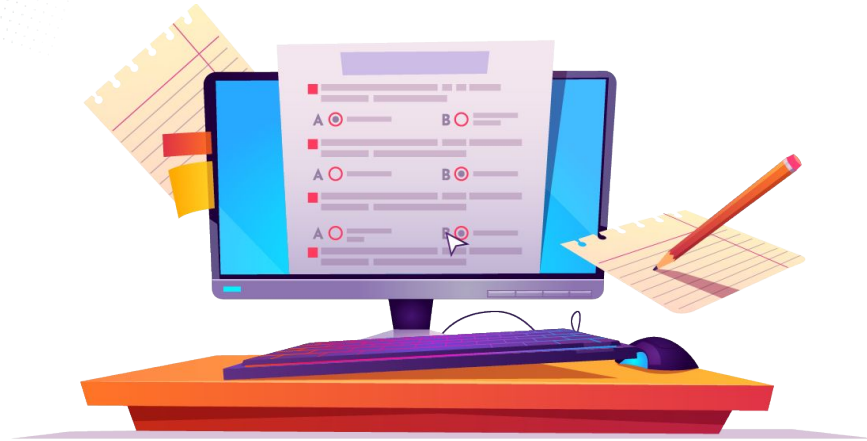
How to Learn a New Programming Language in Record Time ??

Key Takeaways

- ✓ How to Master a New Programming Language
 - Three Key Areas to Focus
- ✓ Programming Languages
 - Types of Programming Languages
- ✓ Choosing a new Programming language
 - Python



Practice



Next Session

Introduction to Python Programming

Feedback





THANK YOU





ALL THE BEST

